

Movie Database Analysis

Agenda:

- **Part 1:** Guided SQL queries
- **Part 2:** IMDB exploratory analysis
- **Deliverables:** data exploration findings, business question and visualization, data cleaning tasks

Key Findings from the IMDB Dataset (2010–2019)

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Data Highlights

- Total movies: [145,081]
- Missing runtime: [30,765] movies
- Missing genre info: [5,359] movies
- Yearly peak: [2017] with [17,504] releases
- Most common genre: [Drama] with [13,803] movies

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Data Scope

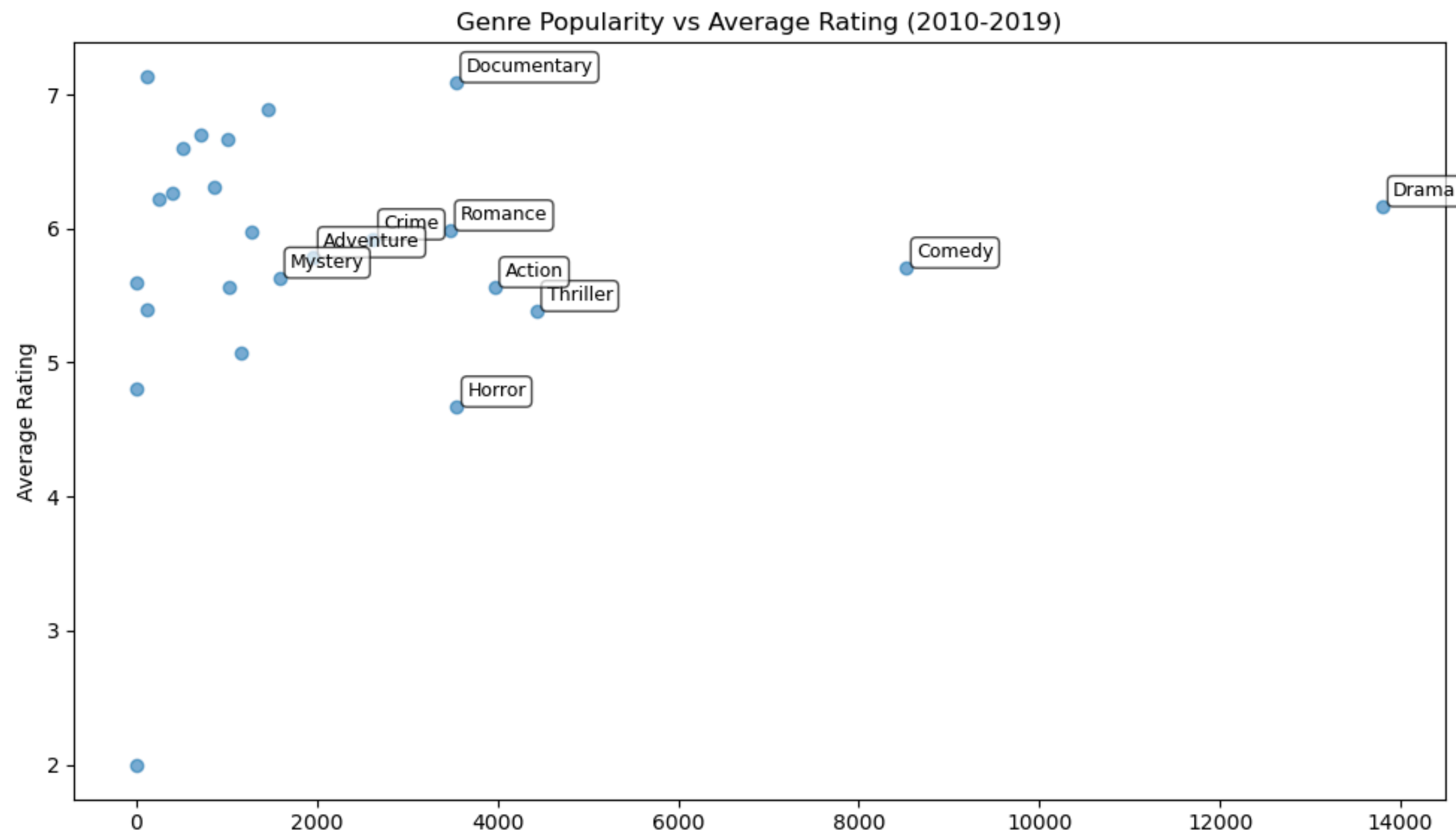
- Highest avg rating (≥ 50 movies): [top genre] at [avg rating]
- Included: genres, runtimes, directors, writers, actors, IMDb ratings
- Excluded: movies with NULL values in critical fields

Dataset: IMDB Movies (2010–2019)

The dataset covers a decade of film production, capturing genre classifications, runtime data, key personnel (directors, writers, actors), and IMDb audience ratings.

Movies with NULL values in critical fields were excluded to ensure analysis reliability. Genre and rating distributions were examined to identify patterns across the full dataset period.

Business Question: Do Movie Genres with More Releases Have Higher Average IMDb Ratings?



[INSERT YOUR SCATTER PLOT IMAGE HERE]

- Popular genres (Drama, Comedy, Action) show high volume but a wide rating spread of 5.5 – 7.5, indicating inconsistent quality at scale.
- Niche genres (Documentary, Film-Noir, War) have fewer releases yet consistently cluster above 6.5 average rating.
- Conclusion: More movies in a genre does NOT guarantee higher ratings — smaller, curated genres can achieve stronger critical reception.

Recommended Cleaning Steps for Reliable Analysis

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Missing Values: runtime_minutes and genres have many nulls – drop, impute, or exclude. **Genres Format:** split comma-separated strings into individual rows and trim spaces. **Vote Threshold:** filter by numvotes $\geq$ 100 to ensure rating reliability.

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Duplicate Checks: verify uniqueness of movie_id across tables. **Date Consistency:** enforce the 2010–2019 window strictly. **Genre Standardisation:** harmonise variations (e.g., "Sci-Fi" vs. "Sci Fi").

Technical Approach

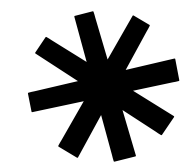
01

Language: Python · Libraries: Pandas, SQLite3, Matplotlib · Data Source: IMDB subset (2010–2019) · Key SQL: JOIN, GROUP BY, COUNT, AVG

02

Data Wrangling: Pandas `.str.split()` and `.explode()` to handle multi-value genres · Code: All analysis in `SQLSummativeLab.ipynb`

Thank You



Presented

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